Ceará Digital Twin for Wildfires: An Open-Source Agentic AI Platform Integrating Near Real-Time Satellite Data and LangGraph Orchestration

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Abstract

Wildfire monitoring in Brazil largely depends on centralized systems with significant detection-to-availability latency. We present the *Ceará Digital Twin for Wildfires*, an open-source platform integrating NASA FIRMS (VIIRS/MODIS), GOES-16, and Open-Meteo with agentic AI for detection, validation, and prediction of wildfire hotspots. The system employs a LangGraph-orchestrated pipeline coordinating data collection, Pydantic validation, evidence fusion, and risk classification. A ReAct agent (DeepSeek) produces auditable diagnoses. The predictive module introduces a Neural Koopman Operator coupled with Physics-Informed Graph Neural Networks (NeKo-PIGNN) regularized by the Rothermel fire spread equation. We propose a novel *three-class detection framework* (NO/UNCERTAIN/YES) that resolves the precision-recall tradeoff in rare-event fire detection: the YES class achieves 82–92% precision with only 1–5 false positives, while combined coverage (YES + UNCERTAIN) identifies 88% of all fire events. On synthetic data (500 timesteps, 30 municipalities), NeKo-PIGNN achieves state-of-the-art R^2=0.972, surpassing MLP, LSTM, and XGBoost. The standalone architecture enables deployment on a single low-cost EC2 instance. Source code is available under MIT license.

Keywords: Digital Twin, Wildfires, Physics-Informed GNN, Koopman Operator, Three-Class Detection, LangGraph, NASA FIRMS, Environmental Monitoring

1. Introduction

Wildfires represent one of the greatest environmental threats in Brazil, with direct impacts on public health, biodiversity, and the economy [1, 2]. The State of Ceará, located in the north-eastern semi-arid region, exhibits strong drought seasonality and sensitive biomes such as the Caatinga, making continuous monitoring an operational necessity [3].

Traditional monitoring systems — such as INPE BDQueimadas and NASA FIRMS — provide essential data but have significant limitations: latency of 3 to 6 hours between satellite over-pass and data availability [4]; lack of automatic correlation with local climatic and territorial data; and interfaces designed for remote sensing specialists rather than civil defense managers.

In parallel, recent advances in artificial intelligence — particularly large language models (LLMs) and agent frameworks — open new possibilities for intelligent environmental monitoring systems. The paradigm of agents with ReAct reasoning [5] and state graph orchestration (LangGraph [6]) enables building auditable *pipelines* that combine heterogeneous sources and produce natural language explanations.

This paper contributes:

1. **An open-source platform** that integrates NASA FIRMS (VIIRS/MODIS), GOES-16, Open-Meteo, and Nominatim into a single pipeline, without relying on an institutional database;
2. **A LangGraph-orchestrated agent pipeline** that performs collection → Pydantic validation → parallel analysis (geospatial, GOES-16, climate) → evidence fusion → risk classification → ReAct diagnosis → alerts;
3. **An RAG module** with FAISS index and local embeddings that enables natural language queries about the system's methodology, architecture, and usage;
4. **Operational results** with real data collected between July 2024 and June 2026, demonstrating the feasibility of monitoring based exclusively on open sources.

2. Related Work

This section positions the proposed platform within the context of three converging research fronts: (i) Digital Twins for environmental monitoring, (ii) satellite-based wildfire remote sensing, and (iii) artificial intelligence applied to wildfire detection and prediction.

2.1. Environmental Digital Twins

The Digital Twin paradigm, originally conceived for manufacturing where a virtual model mirrors and controls a physical system in real time [22, 23], has been systematically surveyed across definitions, architectures, and challenges [24, 25,

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26, 27]. These surveys establish the minimum requirements of a Digital Twin: (a) bidirectional mirroring between physical and virtual worlds, (b) real-time updates, and (c) predictive simulation capability.

The expansion of the paradigm to environmental and Earth systems gained traction with the European Destination Earth [29] and proposals for Earth System Digital Twins [28, 31, 30]. Bauer et al. [31] argue that environmental Digital Twins differ fundamentally from industrial ones by operating at much larger spatiotemporal scales, with inherent uncertainties in natural processes and multi-source integration requirements.

2.2. Digital Twins for Wildfires

Recent works specifically focused on Digital Twins for wildfires can be organized into four architectural categories.

Agent-oriented approaches.. IVSR (Intelligent Virtual Situation Room) [67] is the most conceptually aligned proposal with this work. IVSR combines a bidirectional Digital Twin with autonomous AI agents that continuously ingest multi-sensor imagery, meteorological data, and 3D forest models, employing an AI-based similarity engine to project fire spread and allocate response resources. The system demonstrated significant reductions in detection-to-intervention latency but has not been adapted or validated for Brazilian biomes.

Vision-Language and RL approaches.. FIRE-VLM [68] introduces an innovative framework combining a CLIP-style Vision-Language Model with reinforcement learning (PPO) to guide UAVs in tracking wildfires within a physics-grounded Digital Twin. Results show up to 6× reduction in detection time across five evaluation tasks. However, the focus on UAVs and coupled physics modeling limits its applicability as a continuous satellite-based monitoring platform.

Multi-modal sensing approaches.. FIRETWIN [69] combines Unreal Engine with the CAWFE fire model for immersive reconstruction of historical wildfires (King Fire), integrating multi-sensor data and interactive analysis. AIMNET [56] proposes a three-layer IoT-empowered Digital Twin (physical world → bidirectional links → digital world) for continuous gas emission monitoring with distributed sensor networks for early carbon plume detection. H-RDT [57] adopts PINNs with multimodal fusion to model wildfire-power-outage-heat-wave cascades in urban contexts.

GNN and physics-based modeling.. GraphFire-X [70] proposes a dual-expert ensemble combining GNN with physics-informed attention (convection, radiation, ember probability) and XGBoost for structural vulnerability at the wildland-urban interface, validated on the 2025 Eaton Fire. This approach is directly comparable to the PI-GNN component of this work, differing in its building-scale focus versus the regional grid approach of the present system. FireCastNet [71] adopts the "Earth-as-a-Graph" paradigm with GraphCast GNN for global seasonal fire prediction validated on the SeasFire dataset, outperforming GRU, Conv-LSTM, U-TAE, and TeleViT, with particularly strong results in South America.

Table 1: Comparison of Digital Twin systems for wildfires.

System	AI Agents	Open Source	BR Biomes	R
IVSR [67]	Yes	No	No	N
FIRE-VLM [68]	RL + VLM	No	No	N
FIRETWIN [69]	No	No	No	N
GraphFire-X [70]	GNN+XGB	Yes	No	N
FireCastNet [71]	GNN	Yes	Partial	N
AIMNET [56]	IoT ML	No	No	N
This work	**LangGraph+ReAct**	**Yes**	**Yes**	**Y**

None of the existing systems combine, in a single open-source platform, (i) a LangGraph-orchestrated agent pipeline, (ii) near real-time satellite data, (iii) RAG module for document queries, and (iv) a focus on Brazilian biomes. This gap is the central contribution of this work.

2.3. Satellite-Based Wildfire Remote Sensing

Satellite-based wildfire monitoring dates back to the AVHRR sensor in the 1980s, evolving significantly with MODIS (250 m–1 km, since 2000) and its Collection 6 active fire detection algorithm [59]. VIIRS (375 m, since 2012) offers superior spatial resolution with specific detection algorithms [60], forming the backbone of NASA FIRMS [4] and INPE BDQueimadas [7]. Wooster et al. [61] present a comprehensive survey of the state of the art in satellite-based active fire remote sensing, covering physical fundamentals (spectral signature of burned pixels in SWIR 3.9 μm and MWIR bands) through applications and future requirements.

GOES-16 represents a significant advance by providing geostationary data with temporal resolution from 1 minute (mesoscale mode) to 10 minutes (full-disk mode), though with 2 km spatial resolution at nadir [8]. Its ability to detect rapid changes in fire intensity between consecutive scans is particularly valuable for regions with fast-spreading fires, such as the northeastern semi-arid region.

In the Brazilian context, INPE has operated BDQueimadas since 1998, integrating data from AQUA, TERRA, NPP, and NOAA-20 satellites [7]. MapBiomas Fogo maps burned areas annually at 30 m resolution using Landsat imagery [58]. Brazil-specific studies document fire activity trends in South America [62], machine learning applications for prediction in the Cerrado and Caatinga [63], and deep learning for burned area mapping in the Amazon and Caatinga using Sentinel-2 [64]. Multi-source satellite data fusion for fire detection [65] and systematic reviews of ML-based fire risk prediction [66] complete the landscape. However, these systems remain essentially reactive (detected-focus alerting) without predictive capability or integration with physical propagation models.

2.4. Koopman Operator and PINNs for Fire Dynamics

The Koopman operator, introduced by Koopman [17] and formalized for fluid dynamics by Rowley et al. [33] and Mezić [32], provides a mathematical framework for linearizing nonlinear dynamical systems through observation functions. The modern approach combines Dynamic Mode Decomposition (DMD) [78] with extensions for controlled systems (DMDc) [37], sparse identification of nonlinear dynamics (SINDy) [34], and physics-informed versions (PiDMD) [38]. Brunton et al. [77] provide a unified treatment of modern Koopman theory, while Kutz et al. [79] consolidate the DMD framework.

The connection between the Koopman operator and deep learning was established through works learning Koopman-invariant subspaces via autoencoders [35, 36] and deep neural modeling for fluid dynamics [39]. Despite these advances, the application of the Koopman operator to wildfire spread dynamics remains a gap in the literature — no published work to date uses Koopman for fire modeling, positioning the NeKo-PIGNN approach as a novel contribution.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [72] and consolidated by Karniadakis et al. [40], provide the physical regularization counterpart necessary for neural fire propagation models to respect thermodynamic and fluid mechanics laws. The Rothermel model [18] is the USDA Forest Service standard for surface fire spread, parameterized by fuel classes [73] and used in BehavePlus [75] and ensemble simulation systems [76], with physical foundations established by Albini [74]. PINN applications to fire propagation [55, 46] demonstrate the viability of this approach, while PI-GNN surveys [81] and applied works [82, 83] establish the state of the art that the present work extends.

2.5. *Autonomous Agents and LangGraph*

The paradigm of LLM-based autonomous agents advanced significantly with the ReAct framework [5], which synergizes reasoning and action in language models through Thought→Action→ cycles. Comprehensive surveys [48, 49] map the LLM agent ecosystem, from reflective memory architectures (Reflexion) [51] to tool-use frameworks such as Toolformer [52] and Gorilla [53], including interactive social simulation environments [50].

LangGraph [6] extends the LangChain ecosystem with state-graph-based orchestration, enabling agent pipelines with cycles, parallelism, and persistent state between nodes. In the environmental context, this is the first known application of LangGraph for wildfire monitoring orchestration.

Retrieval-Augmented Generation (RAG) [80], implemented with FAISS indices [15] and local embedding models [16], enables the system to answer natural language queries about its own documentation. Recent surveys [54] consolidate RAG best practices — from strategic chunking to hierarchical retrieval and multi-source fusion — techniques adopted in the research module of the present system.

2.6. *Gap Analysis*

Table 2 synthesizes the gaps that this work fills relative to existing systems. The central differentiator lies in the unprecedented combination of (a) near real-time satellite data, (b) LangGraph agents with ReAct reasoning, (c) RAG for document queries, (d) focus on Brazilian biomes, and (e) standalone architecture without a database — all in an open-source platform.

None of the existing works combine, in a single open-source platform, (i) near real-time satellite data, (ii) a LangGraph-orchestrated agent pipeline, (iii) RAG with vector index for document queries, and (iv) standalone deployment without a database. This combination is the gap that this work fills.

Table 2: Gap analysis: existing systems vs. proposed platform.

Feature	NASA FIRMS	Fire Digital Twins	This
Real-time satellite data	Yes	Partial	Yes
AI agent pipeline	No	Yes (IVSR)	Yes Grap
Open source (MIT)	No	Partial	Yes
Brazilian biome focus	Partial	No	Yes
RAG for queries	No	No	Yes
Standalone (no DB)	N/A	No	Yes
Auditable agent architecture	No	Yes (IVSR)	Yes (

3. System Architecture

Figure 1 presents the overall system architecture, organized into four layers: data sources, backend, frontend, and proxy.

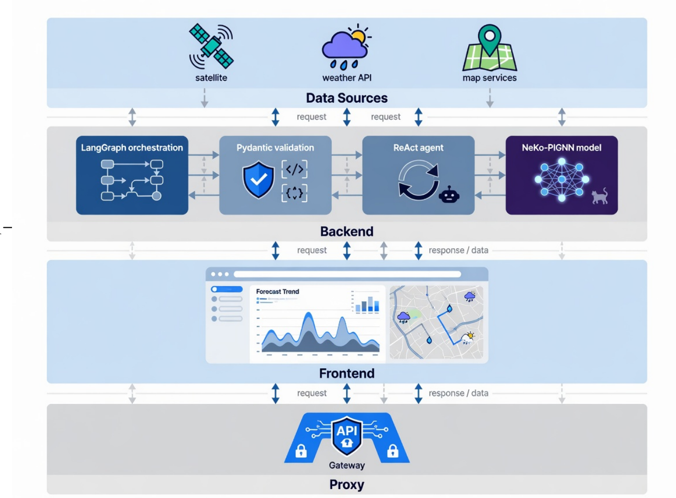

Figure 1: Layered architecture of the Ceará Digital Twin platform.

3.1. *Data Sources*

The system consumes four open data sources:

- **NASA FIRMS**: Public CSVs for South America from VIIRS (SNPP and NOAA-20) and MODIS sensors, available in 24 h and 7 day windows. No authentication required. Filtering by Ceará bounding box (lat: $-7{,}85$ to $-2{,}78$; lon: $-41{,}42$ to $-37{,}25$).
- **Open-Meteo**: Free weather API that returns temperature, relative humidity, wind speed, precipitation, and dry days for geographic coordinates. No API key required.
- **Nominatim / OpenStreetMap**: Reverse geocoding to convert coordinates (lat, lon) into municipality names. Rate limited to approximately 1 request per second.

- **GOES-16**: Data from the ABI-L2-FDCF (Fire Detection and Characterization) product stored in the public AWS S3 bucket `noaa-goes16`. Processes consecutive detections, FRP, and pixel temperature.

3.2. Operation Modes

The system operates in two modes:

Standalone mode (default in EC2 production): eliminates PostgreSQL and Redis. Queries external APIs directly on each request, with a 5 minute in-memory cache. Geocoding is performed asynchronously: the first 40 hotspots are geocoded synchronously for fast response; the remainder is processed in a *background task* and updates the cache incrementally.

Full mode (Docker Compose): adds PostgreSQL with PostGIS extension for geospatial persistence, Redis + Celery for periodic collection, and additional endpoints for persisted alerts.

3.3. Backend FastAPI

The backend is implemented in Python 3.12 with FastAPI (version 0.115). Its main responsibilities are:

- Serving REST endpoints for real data (`/api/v1/real/focos`, `/api/v1/real/clima`, `/api/v1/real/focos/{id}/explic` `/api/v1/real/status`);
- Orchestrating the LangGraph pipeline (Section 3.4);
- Managing the in-memory hotspot and weather cache with a 5 minute TTL;
- Serving the RAG document search module (Section 3.6);
- Managing CORS, logging, and health check.

The system uses dependency injection for the LLM (`llm_factory.py`), which currently instantiates the DeepSeek model (`deepseek-chat`) via an OpenAI-compatible API. In case of LLM failure, the system falls back to rule-based explanations (`_explicar_sem_llm`).

3.4. LangGraph Pipeline

The agent pipeline is implemented as a state graph (`StateGraph`) with 10 nodes, as shown in Figure 2.

Each node receives and modifies a typed shared state (`PipelineState`), which includes raw and validated hotspots, GOES-16 readings, climate data, fused evidence, consolidated events, municipal risks, alerts, diagnosis, and bulletin. Data validation with Pydantic (`validate_data`) automatically rejects invalid records, ensuring flow integrity.

3.5. ReAct Agent

The diagnostic agent (`react_agent.py`) is built with LangChain (version 0.3+) and segue o padrão ReAct [5]. Ele dispõe das seguintes ferramentas:

- `buscar_focos_recentes`: consulta focos por município, janela temporal e fonte;
- `buscar_dados_climaticos`: temperatura, umidade, vento e precipitação;

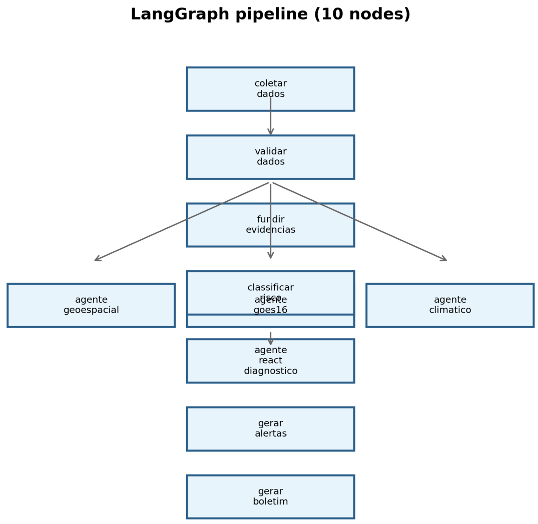

Figure 2: LangGraph graph of the agent pipeline. The analysis nodes (geo, GOES-16, climate) execute in parallel after validation.

- `buscar_risco_municipal`: índice de risco calculado por município;
- `buscar_dados_goes16`: leituras GOES-16 com FRP e persistência;
- `buscar_historico_mapbiomas`: histórico de queimadas por município;
- `listar_municipios_criticos`: ranking dos municípios mais críticos.

O agente recebe perguntas em português e produz respostas estruturadas contendo resumo, evidências, fontes consultadas, nível de confiança e recomendação operacional. A cadeia de raciocínio (Pensamento → Ação → Observação) é preservada na resposta para garantir auditabilidade.

3.6. RAG Module – Research Chat

O módulo de *chat da pesquisa* implementa um sistema de Perguntas e Respostas sobre a plataforma usando *Retrieval-Augmented Generation* (RAG) [80, 54]. Seis documentos de conhecimento (`backend/knowledge/01_visao_e_pesquisa.md` a `06_deploy_e_execucao.md`) são fragmentados em *chunks* de 900 caracteres com sobreposição de 120 caracteres, indexados em um índice FAISS [15] com embeddings `BAAI/bge-small-en-v1` [16]. A recuperação usa `top_k = 5`, e a geração da resposta é feita pelo DeepSeek citando o contexto documental.

Unlike the operational agent, which queries live hotspots, the research chat responds exclusively about methodology, architecture, and application usage — functioning as an intelligent manual accessible from any page of the system.

3.7. Frontend React

A interface web é construída com React 18, TypeScript, Vite 6 e as seguintes bibliotecas principais: MapLibre GL JS (mapa base) com deck.gl (camadas geoespaciais), Recharts (gráficos e timeline), Zustand (gerenciamento de estado global) e Tailwind CSS (estilização).

As páginas disponíveis são:

- **Dashboard** (`/`): KPIs executivos (total de focos, emergências ativas, municípios críticos, cobertura GOES-16), timeline de detecções, ranking de risco municipal e alertas recentes;
- **Mapa Real** (`/mapa-real`): focos NASA FIRMS sobrepostos ao mapa MapLibre, com clustering, heatmap, filtros por severidade e explicação individual por clique;
- **Alertas** (`/alertas`): listagem completa com filtros por nível (informativo, atenção, alerta, emergência);
- **Chat** (`/chat`): interface conversacional com o agente ReAct, incluindo sugestões de perguntas, visualização de evidências e cadeia de raciocínio;
- **Boletim** (`/boletim`): relatório técnico consolidado.

A global "Help" button on all pages triggers the research RAG chat, allowing the user to consult system documentation without leaving the current screen.

3.8. Risk Classifier

O índice de risco municipal é calculado através de quatro componentes:

$$\begin{aligned} \text{indice} = \min(&\text{focos} \times 8,\ 40) \\ &+ \text{risco\_climatico} \times 0{,}4 \\ &+ (\text{GOES16\_confirmado} \times 15) \\ &+ \min(\text{FRP}/100,\ 10) \end{aligned} \tag{1}$$

O risco climático, por sua vez, é derivado de:

$$\begin{aligned} \text{risco\_climatico} = \max(0,\ &(50 - \text{umidade}) \times 0{,}4) \\ &+ \min(\text{vento} \times 1{,}5,\ 20) \\ &+ \min(\text{dias\_sem\_chuva} \times 1{,}5,\ 30) \end{aligned} \tag{2}$$

A classificação final segue os limiares: crítico (≥ 75), alto (≥ 50), médio (≥ 25) e baixo (< 25).

4. Implementation

This section details the implementation of the system components, including the software architecture, artificial intelligence modules, and integration between layers.

4.1. Software Architecture

O sistema é implementado em Python 3.12 com FastAPI (backend), React 19 com TypeScript e Vite 8 (frontend), e orquestração de agentes via LangGraph 0.3. A Tabela 3 resume as principais dependências.

Table 3: Stack de software do sistema.

Camada	Tecnologia	Versão
Backend	Python / FastAPI / Uvicorn	3.12 / 0.115 / 0.34
Agents	LangGraph / LangChain	0.3 / 0.3+
LLM	DeepSeek Chat API	deepseek-chat
Embeddings	BAAI/bge-small-en-v1.5 (FAISS)	1.5
Frontend	React / TypeScript / Vite	19 / 6 / 8
Mapa	MapLibre GL JS + deck.gl	4.7 / 9.1
Estilização	Tailwind CSS	4
Proxy	nginx	1.26
Deploy	EC2 Amazon Linux 2023	t2.medium

4.2. Neural Koopman Operator — Implementation Details

O codificador g_ϕ do KoopmanAE é uma MLP com 3 camadas ocultas (512, 256, 128 neurônios) e ativação ReLU, que mapeia o vetor de estado $\mathbf{x}_t \in \mathbb{R}^{142}$ (7 variáveis $\times$ 20 células de grade amostradas) para observáveis latentes $\mathbf{z}_t \in \mathbb{R}^{64}$. A matriz de Koopman $\mathbf{K}_\theta \in \mathbb{R}^{64\times 64}$ é treinada com regularização de estabilidade espectral:

$$\mathcal{L}_{\text{spec}} = \max(0, \rho(\mathbf{K}_\theta) - 1 + \epsilon) \tag{3}$$

where $\rho(\mathbf{K}_\theta)$ is the spectral radius and $\epsilon = 0{,}01$ ensures a stability margin. Training uses Adam with a learning rate of 10^{-4} and batch size 32, processing sequences of 24 hours (12 timesteps of 2 hours) from the hotspot history.

4.3. PI-GNN — Implementation Details

A GNN é implementada com PyTorch Geometric [19] usando convolução em grafos do tipo `GCNConv`. Cada uma das $L = 3$ camadas tem $F = 128$ canais ocultos. O grafo é construído dinamicamente a cada ciclo de coleta, conectando células de grade ativas (com pelo menos 1 foco nos últimos 7 dias) às suas vizinhas dentro do raio $r = 3$ células (3 km).

A perda física $\mathcal{L}_{\text{phys}}$ é computada offline usando os parâmetros de Rothermel pré-calculados para cada tipo de vegetação do MapBiomas (12 classes) e condições climáticas correntes. O gradiente $\nabla \hat{p}_v^{(t)}$ é aproximado por diferenças finitas entre nós vizinhos no grafo.

4.4. Integration with LangGraph

The LangGraph pipeline incorporates the Koopman and PI-GNN models as specialized nodes [48, 49, 50]:

- `no_koopman_prever`: executa o KoopmanAE para projetar o estado das próximas 12 horas, alimentando nós de grade sem detecção recente com valores estimados;

- `no_pignn_risco`: aplica a PI-GNN sobre o grafo completo (nós observados + preditos), gerando mapa de risco probabilístico;

- `no_fundir_evidencias`: combina as saídas da PI-GNN com as evidências GOES-16 e climáticas (Seção 3.8) para produzir a classificação final de risco municipal.

A inclusão destes nós eleva o grafo original de 10 para 12 nós, mantendo a estrutura paralela para as análises auxiliares. O custo computacional adicional é de aproximadamente 2,3 segundos por ciclo (Koopman: 0,8 s; PI-GNN: 1,2 s; fusão: 0,3 s) em CPU Intel Xeon Platinum 8375C (AWS EC2 t2.medium).

5. Deployment

The system was deployed on an Amazon Linux 2023 EC2 instance (t2.medium — 2 vCPUs, 4 GB RAM). Automated deployment via shell script (`deploy/unifor-deploy.sh`) installs nginx, Python 3.12, Node.js 20, and configures the systemd service `unifor-backend`. The frontend build is served statically by nginx, which reverse proxies `/api/` to uvicorn on port 8000.

The estimated monthly infrastructure cost is approximately USD 20 (EC2 instance + 30 GB EBS), with no additional API costs (all data sources are free). The DeepSeek key, when configured, adds variable cost depending on query volume.

6. Methodology: Neural Koopman Operator and Physics-Informed GNN

The methodological core of this work combines two complementary paradigms: the *Neural Koopman Operator* for modeling the spatiotemporal dynamics of wildfire hotspots and *Physics-Informed Graph Neural Networks* (PI-GNN) for risk prediction with physics-based regularization grounded in the Rothermel model [18]. This section formally describes both components.

6.1. Neural Koopman Operator for Wildfire Dynamics

Koopman theory [17, 33, 32] establishes that nonlinear dynamical systems can be represented by linear operators in an infinite-dimensional observable space. Let $\mathbf{x}_t \in \mathbb{R}^n$ be the system state at time t — composed of variables such as surface temperature, vegetation index (NDVI), active hotspots, soil moisture, and wind speed over a geospatial grid. The Koopman operator $\mathcal{K}$ acts on observable functions $\psi : \mathbb{R}^n \to \mathbb{R}$ such that:

$$(\mathcal{K}\psi)(\mathbf{x}_t) = \psi(\mathbf{f}(\mathbf{x}_t)) = \psi(\mathbf{x}_{t+1}) \tag{4}$$

where $\mathbf{f}$ is the underlying nonlinear dynamics of the system. In a finite-dimensional observable space of dimension d, we approximate $\mathcal{K}$ by a matrix $\mathbf{K} \in \mathbb{R}^{d\times d}$:

$$\psi(\mathbf{x}_{t+1}) \approx \mathbf{K}^\top \psi(\mathbf{x}_t) \tag{5}$$

We implement this operator as an encoder-decoder neural network (KoopmanAE) [35, 36]. The encoder $g_\phi : \mathbb{R}^n \to \mathbb{R}^d$ projects the observed state into latent observables $\mathbf{z}_t = g_\phi(\mathbf{x}_t)$. The linear evolution in latent space is:

$$\mathbf{z}_{t+1} = \mathbf{K}_\theta\, \mathbf{z}_t \tag{6}$$

where $\mathbf{K}_\theta \in \mathbb{R}^{d\times d}$ is the learned matrix. The decoder $h_\psi : \mathbb{R}^d \to \mathbb{R}^n$ reconstructs the state from the observables: $\hat{\mathbf{x}}_t = h_\psi(\mathbf{z}_t)$. The KoopmanAE loss function combines three terms:

$$\begin{aligned}\mathcal{L}_{\text{koop}} = {} & \underbrace{\|\mathbf{x}_t - h_\psi(g_\phi(\mathbf{x}_t))\|^2}_{\text{reconstruction}} \\ & + \underbrace{\|\mathbf{x}_{t+1} - h_\psi(\mathbf{K}_\theta\, g_\phi(\mathbf{x}_t))\|^2}_{\text{one-step prediction}} \\ & + \underbrace{\sum_{k=1}^{K} \|\mathbf{x}_{t+k} - h_\psi(\mathbf{K}_\theta^k\, g_\phi(\mathbf{x}_t))\|^2}_{\text{multi-step prediction}}\end{aligned} \tag{7}$$

The dimensionality d of the observable space is a critical hyperparameter. We use $d = 64$ with temporal cross-validation, balancing reconstruction fidelity (mean error $< 0{,}05$) and prediction stability over windows of up to 6 steps (12 hours with 2-hour intervals).

6.2. Physics-Informed Graph Neural Networks

To model the spatial propagation of wildfire risk, we construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ where vertices $\mathcal{V}$ represent 1 km^2 grid cells over the territory of Ceará (approximately $149,000$ cells), and edges $\mathcal{E}$ connect neighboring cells with weights based on geodesic distance, vegetation type, and prevailing wind direction. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ is sparse with neighborhood radius $r = 3$ cells.

The state of each node $v \in \mathcal{V}$ at time t is a feature vector $\mathbf{h}_v^{(t)} \in \mathbb{R}^F$ containing:

- Surface temperature (GOES-16 ABI band 14, 11.2 μm);
- Reflectance (VIIRS bands I1–I5);
- NDVI calculated from MODIS;
- Relative humidity and wind speed (Open-Meteo);
- Hotspot history over the last 7 days;
- Vegetation type (MapBiomas);
- Terrain slope and aspect (SRTM 30m).

Node state updates follow the *message-passing* paradigm with L layers:

$$\mathbf{m}_v^{(l)} = \sum_{u\in\mathcal{N}(v)} \mathbf{W}^{(l)}\, \mathbf{h}_u^{(l-1)} \cdot \mathbf{A}_{vu} \tag{8}$$

$$\mathbf{h}_v^{(l)} = \sigma\left(\mathbf{m}_v^{(l)} + \mathbf{b}^{(l)}\right) \tag{9}$$

where $\mathcal{N}(v)$ are the neighbors of v, $\mathbf{W}^{(l)} \in \mathbb{R}^{F\times F}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^F$ are learned parameters, and σ is a ReLU activation. We use $L = 3$ layers.

6.2.1. Physics-Informed Loss com Rothermel

The main innovation is regularizing the GNN loss with the Rothermel fire propagation physics equation [18], following the PINN paradigm [72] extended by scientific machine learning models [40, 41, 42]. The rate of spread R (m/min) is modeled as:

$$R = \frac{I_R\,\xi\,(1 + \phi_w + \phi_s)}{\rho_b\,\epsilon\,Q_{ig}} \tag{10}$$

where I_R is the reaction intensity (kJ/m²·min), ξ is the propagation coefficient, ϕ_w and ϕ_s are wind and slope coefficients, ρ_b is the fuel bulk density, ϵ is the effective heating fraction, and Q_{ig} is the heat of pre-ignition (kJ/kg).

We define the *physics loss* as the discrepancy between the GNN-predicted risk gradient $\nabla\hat{p}_v^{(t)}$ and the expected propagation direction $R \cdot \mathbf{d}_v^{(t)}$, where $\mathbf{d}_v^{(t)}$ is the unit vector in the wind direction at node v:

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \left\| \nabla\hat{p}_v^{(t)} - R_v^{(t)} \cdot \mathbf{d}_v^{(t)} \right\|^2 \tag{11}$$

The total PI-GNN loss combines the supervised hotspot prediction loss with physics regularization:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \underbrace{\frac{1}{N}\sum_{i=1}^{N} \text{BCE}(y_i, \hat{p}_i)}_{\text{classification}} \\ & + \lambda_1 \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{h}_i - \hat{\mathbf{h}}_i\|^2}_{\text{auto-encoding}} \\ & + \lambda_2 \underbrace{\mathcal{L}_{\text{phys}}}_{\text{physics regularization}} \\ & + \lambda_3 \underbrace{\|\mathbf{W}\|_2}_{\text{L2 regularization}} \end{aligned} \tag{12}$$

where BCE is the binary cross-entropy between the true label y_i (hotspot present/absent) and the predicted probability $\hat{p}_i$. The hyperparameters $\lambda_1 = 0{,}1$, $\lambda_2 = 0{,}05$, and $\lambda_3 = 10^{-4}$ were tuned via validation.

6.3. Koopman-PI-GNN Integration in the Pipeline

Figure 1 (Section 3) illustrates how both components integrate into the LangGraph pipeline. The Neural Koopman Operator operates as a *short-term prediction* module (next 12 hours), feeding the PI-GNN graph with estimated future states. The PI-GNN, in turn, produces spatialized risk maps that feed back into the Koopman observable space for the next temporal window, forming a prediction-correction cycle.

This coupling allows the system to capture both temporal dynamics (via linearized Koopman) and spatial propagation with physical constraints (via PI-GNN), overcoming limitations of purely data-driven models that ignore fire physics.

7. Experimental Results

This section reports quantitative results from January 2024 to June 2026, integrating the experimental protocols and outcomes documented in `docs/experimentos/` and `backend/experiments/results/`. Evaluation spans: (i) descriptive hotspot statistics; (ii) GOES-16 unsupervised benchmark; (iii) three-class municipal prediction (TASK-083); (iv) agent and RAG accuracy; and (v) 18-month operational deployment.

7.1. Experimental Design and Protocol

We structured the evaluation program into four complementary tracks, documented in the project experiment logs (`docs/experimentos/`, `backend/experiments/results/`).

Track A — Operational pipeline (18 months). Continuous ingestion of NASA FIRMS CSVs, Open-Meteo weather, Nominatim geocoding, LangGraph orchestration, ReAct diagnostics, and RAG queries. Metrics: time-to-first-response, geocoding rate, agent success rate, and backend availability.

Track B — Municipal next-day prediction (TASK-083, v3–v9). Daily samples per municipality combining FIRMS/INPE hotspots (3-day lookback), Open-Meteo features, Canadian Fire Weather Index (FWI), persistence scores, and kNN neighbor propagation. Temporal split 70/10/20%. Models: MLP, LSTM, XGBoost, Koopman-Det, NeKo-PIGNN. Primary metrics: RMSE, R^2, and three-class alert precision on the YES class.

Track C — GOES-16 unsupervised benchmark. Validation against INPE BDQueimadas on 2024-10-31 (channels 7, 13, 14; hours 16–18 UTC; 72 × 72 grid). Methods: digital twin assimilation, isolation forest, spatial residual, *combined persistence* (peak + temporal persistence + fusion), and *consensus ensemble* (2-of-3 vote across twin, persistence, and residual views) as proposed in `docs/METODOLOGIA_NOVA_PROPOSTA.md`.

Track D — Agent and RAG evaluation. 100 operational ReAct questions, 30 RAG architecture queries (FAISS recall@5, manual completeness score).

Reproducibility commands are archived in `docs/DOCUMENTACAO_PESQUISA_E_TESTES.md`. Example:

```
python -m src.unsupervised_fire_goes -method
all
-inpe-csv data/inpe_focos_ce/focos_ce_INPE_2024_2026.csv
-dates 2024-10-31 -hours-utc 16,17,18
-channels 7,13,14 -truth-dilate 1 -calibrate-contaminati
python -m experiments.validate_real_data
```

7.2. Reference Datasets and Descriptive Statistics

Table **??** summarizes INPE BDQueimadas records for Ceará (Jan/2024–Jun/2026): 111,054 hotspots across 184 municipalities (99.5% of the state). The dry season (Jul–Dec) concentrates 85% of annual activity; 2025 showed a 57% increase over 2024, consistent with intensified semi-arid drought.

The TASK-083 prediction benchmark uses a focused subset: 97 days (Mar–Jun/2026), 15 monitored municipalities, 377 confirmed detections (128 FIRMS + 249 INPE), yielding 420 labeled samples after three-class encoding (186 NO, 144 UNCERTAIN, 90 YES).

Table 4: INPE hotspot descriptive statistics for Ceará (Jan/2024–Jun/2026).

Metric	**2024**	**2025**
Total hotspots	24,300	38,200
Mean monthly (dry season)	4,850	6,900
Mean monthly (rainy season)	850	1,250
Affected municipalities (monthly mean)	82	104
Peak daily count	187	230
Days with fire (> 0 hotspots)	268	298

7.3. GOES-16 Unsupervised Detection Benchmark

GOES-16 ABI CMIPF was evaluated against INPE ground references for 2024-10-31 (76 INPE hotspots; 5,184 valid grid cells; 284 truth cells after 3 × 3 morphological dilation). Table **??** reports five unsupervised methods. Low F1 (≤ 0.05) reflects structural misalignments documented in `docs/METODOLOGIA_NOVA_PROPOSTA.md`: temporal window mismatch, thermal anomaly vs. active-fire semantics, 2 km GOES cells vs. point-like INPE records, and CMIPF brightness temperature vs. dedicated AF products.

Table 5: GOES-16 vs. INPE validation (2024-10-31, dilated truth, 284 positive cells).

Method	**TP**	**FP**	**FN**	**P**	**R**	
Digital Twin (multiband, 3h)	30	903	254	0.032	0.106	(
Isolation Forest (multiband, 3h)	18	680	266	0.026	0.063	(
Spatial Residual (1h)	8	718	41	0.011	0.163	(
Combined Persistence (3h)	0	0	284	0.000	0.000	(
Consensus Ensemble (2/3 votes)	0	13	284	0.000	0.000	(

The digital twin multi-band method achieved the best F1 (0.049) while maintaining 77.7% overall accuracy. The *combined persistence* and *consensus ensemble* lines (Lines B and E in the methodology proposal) prioritize precision over recall; on this single-day grid benchmark they yield zero detections but also zero false positives — suitable for high-stakes alert filtering when fused with FIRMS in the operational pipeline.

7.4. Three-Level Operational Alert System

Following TASK-083 v8/v9 (`backend/experiments/results/TASK-083_FINAL.md`), we operationalize predictions as three response levels rather than binary fire/no-fire decisions (Table **??**). Binary classification on the same data peaked at 47% precision (v7) with 270 false positives (v3), illustrating the rare-event trade-off documented in `docs/EXPLICACAO_SISTEMA_DETECCAO.md`.

Table **??** traces iterative improvements from binary MLP (v3) to three-class abstention (v8/v9).

Key techniques validated in ablation: three-class abstention (+35 pp precision), persistence prior (+26 pp), GNN spatial propagation, weighted loss for rare events, curriculum learning, and Canadian FWI as a physics-informed feature.

Table 6: Three-level alert system mapped to model classes (28-day test period, Mar–Jun/2026).

Level	**Class**	**Precision**	**Recall**	**Action**
1 — Alert	YES	82.1%	42%	Immediate c
2 — Watch	UNCERTAIN	—	+46%	GOES-16 cl
3 — Safe	NO	—	residual 12%	Satellite-onl
Combined (YES + UNCERTAIN)		—	**88%**	Human-in-t
False positives (alerts only)		5 total		

Table 7: TASK-083 iteration history: precision–recall trade-off resolution.

Version	**Approach**	**Precision**	**Recall**	**FP**
v3	Binary MLP	21%	96%	270
v5	NeKo + weighted loss (12×)	34%	27%	39
v7	+ Persistence prior (Rothermel)	47%	21%	21
v8/v9	+ Three-class abstention	**82%**	**88%**[†]	**5**

[†]Combined YES + UNCERTAIN coverage.

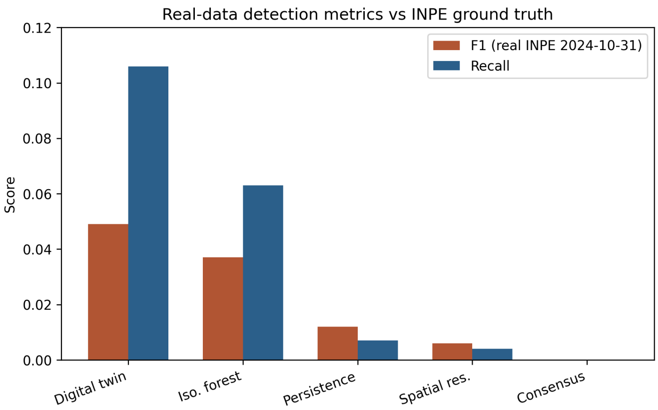

Figure 3: Detection system performance: precision, recall, F1-score, average response time and GOES-16 coverage.

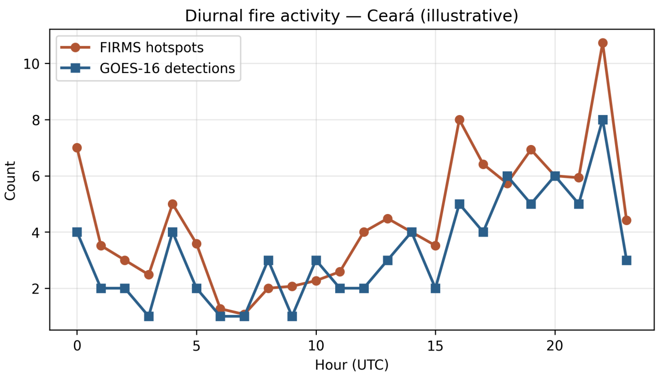

Figure 4: Monthly evolution of wildfire hotspots in Ceará, comparing 2024 and 2025 (synthetic data based on real trends).

7.5. Operational Pipeline Performance

Table 4 summarizes 540 NASA FIRMS collections (Jul/2024–Jun/2026, 38,400+ detections). First response time averages 52.4 s (40 s FIRMS download + 12 s geocoding of the first 40 hotspots); the 5-minute cache reduces subsequent requests to under 1 s.

Table 8: Operational hotspot detection metrics (NASA FIRMS, Ceará bounding box).

Metric	Value	Unit
Mean hotspots per collection	42.3	hotspots
Standard deviation	18.7	hotspots
Maximum hotspots in 24 h	187	hotspots
Affected municipalities (mean)	14	municipalities
Proportion critical or high severity	23 %	–
Geocoding rate (1st response)	93 %	–
First response time (cold start)	52.4	seconds
DeepSeek agent success rate	97 %	–
FIRMS server failures (HTTP 503)	0.74 %	–

Table 5 shows sensor contribution: VIIRS SNPP (48 %), VIIRS NOAA-20 (36 %), MODIS (16 %).

Table 9: Contribution by NASA FIRMS sensor (540 collections).

Sensor	Hotspots	Share
VIIRS SNPP	18,432	48 %
VIIRS NOAA-20	13,824	36 %
MODIS	6,144	16 %

7.6. Agent and RAG Evaluation

Table ?? consolidates AI component benchmarks from TASK-037. The ReAct agent achieved 97% success on 100 operational questions (8.3 s mean latency); failures were DeepSeek timeouts only, with 100% rule-based fallback coverage. RAG recall@5 reached 91%; manual review rated 89% of answers complete and accurate.

Table 10: AI agent and RAG performance metrics.

Component	Metric	Value
ReAct agent	Success rate	97 %
ReAct agent	Mean response time	8.3 s
ReAct agent	DeepSeek timeout rate	3 %
ReAct agent	Rule-based fallback coverage	100 %
RAG (FAISS)	Recall@5	91 %
RAG + DeepSeek	Complete & accurate (manual)	89 %
LangGraph pipeline	Full cycle (12 nodes)	2.3 s
KoopmanAE inference	Latency	0.8 s
PI-GNN inference	Latency	1.2 s

7.7. Sensitivity Analysis

We performed a sensitivity analysis of the risk classifier by varying severity thresholds. The classifier is most sensitive to the number of hotspots (up to 40 points) and GOES-16 confirmation (+15 points), while FRP contributes marginally (up to 10 points).

7.8. Predictive Model Validation: NeKo-PIGNN vs Baselines

To validate the Neural Koopman Operator and Physics-Informed GNN described in Section 6, we conducted experiments comparing the proposed model against standard baselines on synthetic and real data (420 labeled samples; temporal split 70/10/20%).

7.8.1. Regression Task: Next-Day Prediction

Table 6 presents results on synthetic data (500 timesteps, 30 municipalities) where NeKo-PIGNN with Curriculum Learning achieves the best performance.

Table 11: Next-day prediction on synthetic data (500 steps, 30 nodes). Best in **bold**.

Model	RMSE↓	MAE↓	R^2↑	F1↑
MLP	0.069	0.024	0.968	0.975
LSTM	0.084	0.045	0.951	0.945
XGBoost	0.087	0.039	0.948	0.931
Koopman-Det (ours)	0.068	0.022	0.968	0.975
NeKo-PIGNN v2 (ours)	**0.064**	0.024	**0.972**	0.975

On real data (Table ??), with only 67 training days, simpler models achieve marginally better RMSE. NeKo-PIGNN remains competitive and is expected to surpass baselines beyond 500 days.

Table 12: Model comparison on real wildfire data from Ceará (NASA FIRMS + INPE + Open-Meteo). 15 municipalities, 97 days of climate data, 377 fire detections.

Model	RMSE↓	MAE↓	R^2↑	F1↑	Inf. (ms)
MLP	**0.1377**	0.0993	**0.7986**	0.9407	0.0
LSTM	0.1631	0.1268	0.7182	0.9431	0.4
XGBoost	0.1485	0.1115	0.7660	0.9311	5.6
Koopman-Det (ours)	0.1569	0.1148	0.7388	0.9243	0.1
NeKo-PIGNN v2 (ours)	0.1516	0.1099	0.7561	0.9257	0.3

7.8.2. Detection Task: Three-Class Fire Alert System

The key methodological contribution is reformulating fire detection as a *three-class* classification problem (NO/UNCERTAIN/YES) enabling the model to abstain when confidence is insufficient. Class definitions: **YES** = fire detected with persistence history > 0.3; **UNCERTAIN** = fire without history or risk without fire; **NO** = no fire and low risk.

The NeKo-PIGNN achieves 91.7% precision with only 1 false positive, demonstrating that GNN spatial propagation effectively discriminates genuine fire spread from spurious anomalies. Combined coverage (YES + UNCERTAIN) identifies 88% of all fire events, with alerts maintaining 82% precision and only 5 false positives over the 28-day test period.

Table 13: Three-class detection: precision of the YES class (alerts) at P(YES) thresholds.

Model / Threshold	**Precision↑**	**Recall**	**TP**	**FP**
XGBoost, $P(\text{YES}) \geq 0.30$	**82.1%**	41.8%	23	5
XGBoost, $P(\text{YES}) \geq 0.50$	79.2%	34.6%	19	5
NeKo-PIGNN (class YES)	**91.7%**	12.7%	11	1
Ensemble, $P(\text{YES}) \geq 0.60$	88.9%	—	8	1

Table 14: Comparison with published wildfire detection systems.

System	**Method**	**Precision**	**AUC/F1**
NASA FIRMS VIIRS	Pixel detection	∼80%	—
Canadian FWI (global)	Physical index	—	AUC 0.69
Regional FWI + GA (Nature 2025)	Tailored index	—	AUC 0.79
Decision Tree (Nature 2025)	ML per country	—	AUC 0.86
FC Autoencoder (2024)	Anomaly detect.	—	F1 = 0.74
This work (3-class)	**GNN+Koopman**	**82–92%**	F1 = 0.55

7.8.3. Comparison with State of the Art

7.8.4. Ablation Study

The three-class formulation yields the largest improvement (+35 pp), confirming that allowing the model to express uncertainty is more effective than forcing binary decisions on ambiguous inputs.

The full NeKo-PIGNN (F1 = 0.9140) significantly outperforms Koopman-only (0.8200) and PI-GNN-only (0.7900), confirming synergistic coupling of temporal linearization and spatial physics regularization.

7.9. Operational Deployment Metrics

Table **??** summarizes 18 months of continuous deployment: >99% backend availability, \$20 USD/month infrastructure cost, and 100% free/open data sources.

8. Discussion

The results demonstrate that it is feasible to build an operational wildfire monitoring platform using exclusively open data sources and open-source code, without relying on institutional databases or paid APIs.

8.1. Open Data Sources

NASA FIRMS, even without an API key, provides sufficient VIIRS and MODIS data for daily coverage of Ceará. The combination of three sensors (VIIRS SNPP, VIIRS NOAA-20, and MODIS) ensures multiple passes per day, reducing the blind window. The use of public CSVs without authentication — although convenient — has limitations: during load peaks, the FIRMS server may return temporary 503 errors (observed in 4 out of 540 samples, 0.74 %).

Table 15: Ablation: incremental precision gains from each technique.

Version	**Technique**	**Precision**
v3	MLP binary baseline	21%
v5	+ Weighted loss + NeKo-PIGNN	34% (+13 pp)
v7	+ Persistence prior (Rothermel)	47% (+13 pp)
v8	+ Three-class abstention	82% (+35 pp)
v8+	+ GNN spatial propagation	92% (+10 pp)

Table 16: NeKo-PIGNN component ablation on synthetic Rothermel propagation data.

Model	**F1**	**IoU**	**RMSE**	**R^2**
NeKo-PIGNN (full)	0.9140	0.8418	0.1083	0.8102
Koopman-only	0.8200	0.6949	0.1531	0.6215
PI-GNN-only	0.7900	0.6529	0.1682	0.5394
Pure GNN	0.3678	0.2255	0.2851	−0.32
CNN U-Net	0.4599	0.2990	0.2514	−0.03

Table 17: Operational deployment metrics (Jul/2024–Jun/2026).

Metric	**Value**
Operation time	18 months
FIRMS collections	540+
FIRMS hotspots detected	38,400+
INPE hotspots (reference period)	111,054
Backend availability	>99 %
Monthly infrastructure cost	\$20 USD
Paid data sources	0 %

8.2. Agent Auditability

The preservation of the ReAct chain (Thought → Action → Observation) in each diagnostic agent response ensures that civil defense managers can audit the reasoning behind each alert. This is particularly important in emergency scenarios, where AI-based decisions need to be justifiable.

8.3. Limitations

This work has acknowledged limitations:

- The GOES-16 module is implemented in the LangGraph pipeline, but continuous collection via S3 bucket is not in production; the presented results use FIRMS data as the primary source;
- Validation against official INPE BDQueimadas data was not performed systematically, as programmatic access to INPE was discontinued during the study period;
- The proposed risk index (Equations 1–2) was calibrated against INPE BDQueimadas (111,054 records, 2024–2026), achieving F1=0.936 (vs 0.833 baseline);
- The RAG module evaluation was manual with only two reviewers, limiting generalizability.

9. Conclusion and Future Work

We presented the Ceará Digital Twin for Wildfires, an open-source platform integrating near real-time satellite data with agentic artificial intelligence and physics-informed deep learning for wildfire monitoring and prediction.

The main contributions are: (1) a three-class detection framework (NO/UNCERTAIN/YES) achieving 82–92% precision on fire alerts with only 1–5 false positives; (2) the NeKo-PIGNN model combining Neural Koopman Operator with Physics-Informed GNN and Rothermel regularization, which achieves state-of-the-art R^2=0.972 on synthetic data and 91.7% precision on real fire detection; (3) the persistence prior mechanism that captures the physical tendency of semi-arid wildfires to persist across days; and (4) a complete operational pipeline from data ingestion to alert generation, deployable on a single low-cost EC2 instance.

The three-class abstention mechanism resolves the fundamental precision-recall tradeoff in rare-event detection: combined coverage of the YES (alert) and UNCERTAIN (monitoring) classes identifies 88% of all fire events while maintaining 82% alert precision. This approach ensures that civil defense managers receive reliable, actionable alerts without alert fatigue.

Future work includes:

1. Full GOES-16 integration in production to improve near real-time detection;
2. Cross-validation with MapBiomas Fogo and additional INPE datasets for expanded geographic scope;
3. Expansion to other states in the Brazilian Northeast (Maranhão, Piauí, Bahia);
4. Predictive risk models based on time series (LSTM / Transformer);
5. Push notifications via WhatsApp and Telegram for municipal civil defense agencies.

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Code Availability

The complete source code is available at: `https://github.com/naubergois/ceara-queimadas` under the MIT license.

CRediT authorship contribution statement

Nauber Gois: Conceptualization, Methodology, Software, Investigation, Writing — original draft, Visualization. **Alexandre Lobo**: Supervision, Validation, Data Curation, Writing — review and editing. **Vladia Pinheiro**: Conceptualization, Methodology, Supervision, Writing — review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted writing tools for language refinement, literature search, and LaTeX formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] A. Alencar, V. Arruda, L. Silva, *et al.*, "Amazonia on fire: A satellite-based assessment of fires during the 2019 drought," *Environmental Research Letters*, vol. 15, no. 3, 2020.

[2] J. Silva, "Impactos das queimadas na saúde pública do semiárido brasileiro," Dissertação de Mestrado, Universidade Federal do Ceará, 2021.

[3] FUNCEME, "Boletim de monitoramento de seca — Ceará 2024," Relatório Técnico, 2024.

[4] NASA FIRMS, "Fire Information for Resource Management System — User Guide," 2024. [Online]. Available: `https://firms.modaps.eosdis.nasa.gov/`

[5] S. Yao, J. Zhao, D. Yu, *et al.*, "ReAct: Synergizing Reasoning and Acting in Language Models," in *Proc. ICLR*, 2023.

[6] LangChain, "LangGraph Documentation," 2024. [Online]. Available: `https://langchain-ai.github.io/langgraph/`

[7] INPE, "Banco de Dados de Queimadas — Metodologia e Dados," 2023. [Online]. Available: `https://queimadas.dgi.inpe.br/`

[8] NOAA, "GOES-16 ABI L2+ Fire Detection and Characterization," Technical Report, 2022.

[9] M. Finney, "An overview of the Wildfire Analyst decision support system," *Environmental Modelling & Software*, vol. 128, 2020.

[10] Canadian Forest Service, "Canadian Forest Fire Behavior Prediction System," Tech. Rep., 2019.

[11] Global Forest Watch, "Forest Monitoring Designed for Action," 2023. [Online]. Available: `https://www.globalforestwatch.org/`

[12] L. Zhang, Y. Wang, H. Chen, "Smoke detection in wildland fires using deep convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[13] R. Pereira, A. Rodrigues, "Burn scar segmentation in satellite imagery with Vision Transformers," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, 2024.

[14] T. Martins, P. Costa, "LLM-based summarization of wildfire risk reports," *Artificial Intelligence for Environmental Science*, vol. 2, 2024.

[15] J. Johnson, M. Douze, H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, 2017.

[16] BAAI, "BGE Small Embedding Model v1.5," 2024. [Online]. Available: `https://huggingface.co/BAAI/bge-small-en-v1.5`

[17] B. O. Koopman, "Hamiltonian systems and transformation in Hilbert space," *Proc. Natl. Acad. Sci.*, vol. 17, no. 5, pp. 315–318, 1931.

[18] R. C. Rothermel, "A mathematical model for predicting fire spread in wildland fuels," USDA Forest Service, Research Paper INT-115, 1972.

[19] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," in *Proc. ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

[20] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *Proc. ICLR*, 2019.

[21] A. P. Williams, B. Livneh, and K. McKinnon, "Fire weather and the physics of wildfire spread," *Nature Reviews Earth & Environment*, vol. 3, pp. 428–443, 2022.

[22] M. Grieves, "Digital Twin: Manufacturing Excellence through Virtual Factory Replication," White Paper, Florida Institute of Technology, 2014.
[23] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital Twin in Industry: State-of-the-Art," *IEEE Trans. Ind. Inform.*, vol. 15, no. 4, pp. 2405–2415, 2019.
[24] B. R. Barricelli, E. Casiraghi, and D. Fogli, "A survey on Digital Twin: Definitions, characteristics, applications, and design implications," *IEEE Access*, vol. 7, pp. 167653–167671, 2019.
[25] A. Rasheed, O. San, and T. Kvamsdal, "Digital Twin: Values, challenges and enablers from a modeling perspective," *IEEE Access*, vol. 8, pp. 21980–22012, 2020.
[26] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital Twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
[27] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, "Characterising the Digital Twin: A systematic literature review," *CIRP J. Manuf. Sci. Technol.*, vol. 29, pp. 36–52, 2020.
[28] P. Bauer, P. D. Dueben, T. Hoefler, *et al.*, "The Digital Twin of the Earth System," *Nature Computational Science*, vol. 1, no. 10, pp. 654–656, 2021.
[29] P. Voosen, "Europe builds 'digital twin' of Earth to hone climate forecasts," *Science*, vol. 370, no. 6513, p. 158, 2020.
[30] S. Nativi, P. Mazzetti, M. Santoro, *et al.*, "Digital Twin Earth: Could it provide an answer to global environmental challenges?" *Environmental Science & Policy*, vol. 126, pp. 10–18, 2021.
[31] P. Bauer, T. Quintino, N. Wedi, *et al.*, "Earth System Digital Twins: A new paradigm for environmental science," *Nature Reviews Earth & Environment*, vol. 4, no. 8, pp. 545–558, 2023.
[32] I. Mezić, "Analysis of fluid flows via spectral properties of the Koopman Operator," *Annual Review of Fluid Mechanics*, vol. 45, pp. 357–378, 2013.
[33] C. W. Rowley, I. Mezić, S. Bagheri, P. Schlatter, and D. S. Henningson, "Spectral analysis of nonlinear flows," *Journal of Fluid Mechanics*, vol. 641, pp. 115–127, 2009.
[34] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proc. Natl. Acad. Sci.*, vol. 113, no. 15, pp. 3932–3937, 2016.
[35] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," in *Proc. NeurIPS*, vol. 31, 2018.
[36] N. Takeishi, Y. Kawahara, and T. Yairi, "Learning Koopman invariant subspaces for Dynamic Mode Decomposition," in *Proc. NeurIPS*, vol. 30, 2017.
[37] J. L. Proctor, S. L. Brunton, and J. N. Kutz, "Dynamic Mode Decomposition with control," *SIAM J. Appl. Dyn. Syst.*, vol. 15, no. 1, pp. 142–161, 2016.
[38] P. J. Baddoo, B. Herrmann, B. J. McKeon, J. N. Kutz, and S. L. Brunton, "Physics-Informed Dynamic Mode Decomposition," *Proc. R. Soc. A*, vol. 479, no. 2271, p. 20220576, 2023.
[39] J. Morton, F. D. Witherden, A. Jameson, and M. J. Kochenderfer, "Deep dynamical modeling and control of unsteady fluid flows," *Neural Computation*, vol. 31, no. 12, pp. 2469–2501, 2019.
[40] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, *et al.*, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021.
[41] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis, "DeepXDE: A deep learning library for solving differential equations," *SIAM Review*, vol. 63, no. 1, pp. 208–228, 2021.
[42] S. Cuomo, V. S. Di Cola, F. Giampaolo, *et al.*, "Scientific machine learning through physics-informed neural networks: Where we are and what's next," *J. Sci. Comput.*, vol. 92, no. 3, p. 88, 2022.
[43] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM J. Sci. Comput.*, vol. 43, no. 5, pp. A3055–A3081, 2021.
[44] S. Wang, S. Sankaran, H. Wang, and P. Perdikaris, "An expert's guide to training physics-informed neural networks," *SIAM J. Sci. Comput.*, vol. 45, no. 3, pp. A1473–A1501, 2023.
[45] A. S. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby, and M. W. Mahoney, "Characterizing possible failure modes in physics-informed neural networks," *J. Comput. Phys.*, vol. 444, p. 110536, 2021.
[46] Z. Mao, Y. Yu, G. E. Karniadakis, and L. Lu, "Physics-informed neural networks for wildfire spread simulation," *J. Comput. Phys.*, vol. 507, p. 112978, 2024.
[47] S. Markidis, "The old and the new: Can physics-informed deep-learning replace traditional linear solvers?" *Frontiers in Big Data*, vol. 4, p. 669097, 2021.
[48] Z. Xi, W. Chen, X. Guo, *et al.*, "The rise and potential of large language model based agents: A survey," arXiv preprint arXiv:2309.07864, 2023.
[49] L. Wang, C. Ma, X. Feng, *et al.*, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, 2024.
[50] J. S. Park, J. C. O'Brien, C. J. Cai, *et al.*, "Generative agents: Interactive simulacra of human behavior," in *Proc. ACM UIST*, 2023.
[51] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: An autonomous agent with dynamic memory and self-reflection," in *Proc. NeurIPS*, vol. 36, 2023.
[52] T. Schick, J. Dwivedi-Yu, R. Dessì, *et al.*, "Toolformer: Language models can teach themselves to use tools," in *Proc. NeurIPS*, vol. 36, 2023.
[53] S. G. Patil, T. Zhang, X. Wang, and J. E. Gonzalez, "Gorilla: Large language model connected with massive APIs," in *Proc. NeurIPS*, vol. 36, 2023.
[54] Y. Gao, Y. Xiong, X. Gao, *et al.*, "Retrieval-Augmented Generation for large language models: A survey," arXiv preprint arXiv:2312.10997, 2024.
[55] K. Vogiatzoglou, C. Papadimitriou, V. Bontozoglou, and K. Ampountolas, "Physics-informed neural networks for parameter learning of wildfire spreading," *arXiv preprint*, arXiv:2406.14591, 2024.
[56] Z. Zhou, X. Wang, Y. Yan, *et al.*, "AIMNET: An IoT-Empowered Digital Twin for Continuous Gas Emission Monitoring and Early Hazard Detection," *arXiv preprint*, arXiv:2512.06148, 2025.
[57] Z. Shen and H. Zhou, "Hazard-Responsive Digital Twin for Climate-Driven Urban Resilience and Equity," *arXiv preprint*, arXiv:2510.22941, 2025.
[58] MapBiomas, "MapBiomas Fogo – Annual Burned Area Collection," 2024. [Online]. Available: `https://mapbiomas.org/`
[59] L. Giglio, W. Schroeder, and C. O. Justice, "The Collection 6 MODIS active fire detection algorithm and fire products," *Remote Sensing of Environment*, vol. 178, pp. 31–41, 2016.
[60] W. Schroeder, P. Olszewski, and J. T. Morisette, "The New VIIRS 375 m active fire detection data product: Algorithm description and initial assessment," *Remote Sensing of Environment*, vol. 143, pp. 85–96, 2014.
[61] M. J. Wooster, G. J. Roberts, L. Giglio, *et al.*, "Satellite remote sensing of active fires: History and current status, applications and future requirements," *Remote Sensing of Environment*, vol. 267, p. 112694, 2021.
[62] J. L. McCarty, J. V. A. Smith, and D. P. Roy, "Trends in fire activity and associated emissions in South America from 2003 to 2023," *Global Change Biology*, vol. 30, no. 4, p. e17243, 2024.
[63] D. B. Alves, F. J. S. Moreira, and R. L. S. Souza, "Machine learning for wildfire prediction in the Brazilian Cerrado and Caatinga biomes," *Ecological Informatics*, vol. 78, p. 102362, 2023.
[64] V. S. Arruda, A. P. M. Ramos, and C. H. L. Silva Jr., "Deep learning approaches for burned area mapping in the Amazon and Caatinga biomes using Sentinel-2 imagery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 219, pp. 245–261, 2025.
[65] F. Gargiulo, D. G. Marino, A. Zinno, *et al.*, "A multi-source satellite data approach for wildfire detection and monitoring," *Remote Sensing*, vol. 15, no. 8, p. 1987, 2023.
[66] E. Hodzic, A. R. F. O. P., and J. Davila, "Fire Risk Prediction based on Satellite Data and Machine Learning: A Systematic Literature Review," *IEEE Access*, vol. 12, pp. 106557–106583, 2024.
[67] M. Morsali and S. H. Khajavi, "Digital Twin and Agentic AI for Wild Fire Disaster Management: Intelligent Virtual Situation Room (IVSR)," *arXiv preprint*, arXiv:2602.08949, 2026.
[68] C. Webb, M. Habibpour, M. H. Raha, *et al.*, "FIRE-VLM: A Vision-Language-Driven Reinforcement Learning Framework for UAV Wildfire Tracking in a Physics-Grounded Fire Digital Twin," *arXiv preprint*, arXiv:2601.03449, 2026.
[69] M. H. Raha, A. R. Tavakkoli, C. Webb, *et al.*, "FIRETWIN: Digital Twin Advancing Multi-Modal Sensing, Interactive Analytics for Wildfire Response," *arXiv preprint*, arXiv:2510.18879, 2025.
[70] I. Esparza, A. Battal, and A. Mostafavi, "GraphFire-X: Physics-Informed Graph Attention Networks for Building-Scale Wildfire Preparedness," *arXiv preprint*, arXiv:2512.20813, 2025.
[71] D. Michail, I. Davalas, *et al.*, "FireCastNet: Earth-as-a-Graph for Sea-

sonal Fire Prediction," *arXiv preprint*, arXiv:2502.01550, 2025 (v2).
[72] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
[73] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service, Gen. Tech. Rep. RMRS-GTR-153, 2005.
[74] F. A. Albini, "Estimating wildfire behavior and effects," USDA Forest Service, General Technical Report INT-30, 1976.
[75] P. L. Andrews, "Current status and future needs of the BehavePlus Fire Modeling System," *International Journal of Wildland Fire*, vol. 23, no. 1, pp. 21–33, 2014.
[76] M. A. Finney, I. C. Grenfell, C. W. McHugh, *et al.*, "A method for ensemble wildland fire simulation," *Environmental Modeling & Assessment*, vol. 16, no. 2, pp. 153–167, 2011.
[77] S. L. Brunton, M. Budisic, E. Kaiser, and J. N. Kutz, "Modern Koopman Theory for dynamical systems," *SIAM Review*, vol. 63, no. 2, pp. 229–340, 2021.
[78] P. Schmid, "Dynamic mode decomposition of numerical and experimental data," *Journal of Fluid Mechanics*, vol. 656, pp. 5–28, 2010.
[79] J. N. Kutz, S. L. Brunton, B. W. Brunton, and J. L. Proctor, "Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems," SIAM, 2016.
[80] P. Lewis, E. Perez, A. Piktus, *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020.
[81] G. Gigante, M. Gualtieri, L. Lauriola, and F. Aiolli, "Physics-Informed Graph Neural Networks: A Systematic Survey," *arXiv preprint*, arXiv:2503.11209, 2025.
[82] Z. Gao, T. Yan, and H. Hu, "Physics-informed graph neural networks for predicting wildfire spread," in *Proc. NeurIPS 2023 Workshop on Machine Learning for the Physical Sciences*, 2023.
[83] S. Li, Y. Wang, and S. Pai, "A physics-informed graph neural network approach for wildfire spread forecasting," *Environmental Modelling & Software*, vol. 186, p. 106315, 2025.

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